ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CAS5821 MATHEMATICAL FOUNDATIONS OF DATA SCIENCE

MID TERM TEST

DATE: 23RD APRIL 2026

TIME: 17:30 HRS

TIME ALLOWED : ONE AND HALF HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Answer any **TWO** questions from the three questions provided on the question paper. All questions carry 25 marks.
2. Candidates must **not turn this page** until the invigilator tells them to do so.

Question 1

(a) A store is doing inventory for one week to determine the popularity of a new product

Day Number	1	2	3	4	5	6	7
Number sold MB	24	32	41	50	60	68	77

i) Find the *linear regression equation* that models this data. (Round to *nearest integer.)*
Choose:

$y = 8x + 14$
$y = 9x + 15$
$y = 15x + 9$
$y = 14x + 8$ [15 Marks]

ii) If this pattern of growth continues, predict the number sold on the 10^{th}day.
Choose:

94 105 148 159

[10 Marks]

(25 marks)

Question 2

Consider the summary output below:

SUMMARY OUTPUT

Regression Statistics	
Multiple R	0.67565694
R Square	0.45651229
Adjusted R Square	0.35769635
Standard Error	0.48959527
Observations	66

ANOVA

	df	*SS*	*MS*	*F*	*Significance F*
Regression	10	11.07388	1.107388	4.619824	9.07E-05
Residual	55	13.18369	0.239704		
Total	65	24.25758			

a) What is the meaning of the coefficient of determination (R Square) above?

[10 Marks]

b) Draw conclusions for the above output to determine whether the independent variables have an effect on the dependent variable. [15 Marks]

(25 marks)

Question 3

An engineer is interested in studying the relationship between the temperature and resistance of vacuum transducer bobbins, which are used in the automobile industry. She believes that this relationship can be studied in a linear regression model, $Y = \beta_0 + \beta_1 x$ where x is temperature and y is the resistance. After collecting data, she used Minitab to estimate the regression model. Below is the partial Minitab output:

Regression Analysis: Resistance Versus Temperature					
Analysis of variance					
Source	DF	SS	MS	F	P
Regression	1	624.70	624.703	156.91	0.000
Error	22	87.59	?		
Total	?	712.29			

Coefficients				
Predictor	Coef	SE Coef	T	P
Constant	?	4.55	2.83	0.010
Temparature	?	0.06	12.60	0.000

(a) What is the expected value of resistance when the temperature is 70 degrees? [10 Marks]

(b) What is coefficient of determination, r^2? [15 Marks]

(25 marks)

END OF TEST